Luke F. Miller

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Portfolio | LinkedIn

Economist and Ph.D. Candidate

Education

Ph.D. Candidate in Economics, University of Hawai'i at Mānoa — Expected 2027
M.A. Economics, University of Hawai'i at Mānoa — 2023
Master of International Development, University of Pittsburgh — 2020
B.S. Economics, University of Pittsburgh — 2017

Technical Skills

Evaluation and analysis: Causal inference; econometric modeling; randomized controlled trials; instrumental variables; Bayesian structural time series; panel and survey data analysis; quantitative and qualitative research; financial forecasting; rate and cost-of-service analysis.
Data and reporting: Multi-source data integration; quality assurance; coding and automation; data visualization; technical writing; stakeholder communication.
Software: R/RStudio, Excel/VBA, Stata, SQL, Power BI, Tableau, ArcGIS, SAS, LaTeX, Python.
Languages: English; Spanish proficiency.

Relevant Professional Experience

Rate Analyst — July 2025–Present
ElectriCities of North Carolina

- Led annual retail and wholesale revenue forecasting across 19 municipal electric systems, working with city managers, finance directors, and utility leaders to support financial planning.
- Modeled load growth, capital investment requirements, and long-term revenue implications for industrial recruitment and economic development projects.
- Assisted with market energy settlements and invoicing for surplus wholesale power transactions between North Carolina Municipal Power Agency Number 1 (NCMPA1) and external counterparties.
- Supported cost-of-service studies and rate design to equitably allocate infrastructure and energy supply costs; conducted quality assurance audits of retail billing systems.
- Developed VBA automation and data integration tools for forecasting, billing, and reporting, reducing manual compilation and improving consistency across utilities.

Energy Economics Specialist — May 2023–September 2024
Hawai'i State Energy Office

- Led economic analysis and rate forecasting for CERAP 2.0 on Moloka'i, a community-led energy planning effort addressing resilience and net-zero goals.
- Built Excel/VBA models of revenue requirements, generation costs, and customer rate impacts to inform energy planning and infrastructure decisions.
- Secured $395,000 in national laboratory support through a successful grant application to evaluate pumped storage hydropower and its role in Hawaii's energy system.
- Led State Emergency Support Function 12 for energy during the Maui wildfires, coordinating with federal and local agencies to support continuity of energy resources.
- Managed a contract with Honolulu's Department of Planning and Permitting to improve solar and storage permitting and address backlogs and FEMA flood-zone issues.

Certifications

Google Advanced Data Analytics Professional Certificate 2025
Google Data Analytics Professional Certificate 2025
FEMA Professional Development Series, Emergency Management Institute 2020

Select Research Papers and Evaluations

WOW Factor: Tourism Transport Disruption and Destination Impacts in Iceland
Revise and resubmit, Tourism Planning & Development.

- Uses Bayesian structural time-series methods to estimate counterfactual passenger flows following WOW Air's 2019 bankruptcy and examine tourism-related labor-market outcomes.
- Integrates multiple data sources and uses placebo checks and market-level comparisons to assess robustness and document limitations in attribution.

Who's in the Room? Causal Evidence of Political Response Distortion in Surveys

- Uses Eurobarometer survey data and an instrumental-variable approach to investigate third-party presence, response patterns, and demographic heterogeneity.

Additional Relevant Experience

Lecturer and Graduate Teaching Assistant 2021–2024 (intermittent)
University of Hawai'i at Mānoa

- Taught Development Economics (Summer 2024) and Introduction to Statistics (Summer 2022), developing course materials and applied assignments.
- As a Graduate Teaching Assistant (August 2021–May 2023), provided individual tutoring and graded assignments for more than 100 students in macroeconomics and statistics.

Research Fellow August 2022–May 2023
University of Hawai'i at Mānoa / NOAA Sea Grant

- Constructed a 40-year longitudinal climate and economic dataset by integrating multiple sources; applied probit and logit models to examine local flood events and economic behavior.

Non-Profit Consultant January–April 2020
University of Pittsburgh Non-Profit Clinic

- Conducted research and developed recommendations for Alpha House, a therapeutic community serving people with substance use disorders, to inform evidence-based program design.

Entrepreneurial Development & Research Intern May–July 2019
Bright Kids Uganda, Entebbe

- Conducted a needs assessment through stakeholder interviews and introduced basic accounting practices to support local microloan recipients.

Disaster Workforce Engagement Intern January–May 2019
American Red Cross, Southwestern Pennsylvania

- Supported volunteer responder readiness through training and exercises; improved systems for organizing trained volunteers and their response capabilities.

AmeriCorps Project Lead August 2017–July 2018
Habitat for Humanity Vail Valley, Avon, CO

- Managed and trained volunteers supporting the simultaneous construction of six homes, with responsibility for safety and quality oversight.